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Bioptic indicia reader assembly

Abstract

An example bioptic indicia reader assembly includes: a housing having an upper housing portion and a generally upright window positioned in the upper housing portion; and a platter having a proximal edge adjacent the upper housing portion. The upper housing portion comprises a first protrusion located on a first side of a longitudinal centerline of the bioptic indicia reader assembly and a second protrusion located on a second side of the longitudinal centerline. The first protrusion comprises a first camera having a first field-of-view (FOV) having an inner first lateral boundary that extends behind and at an angle to the proximal edge of the platter and the second protrusion comprises a second camera having a second FOV having an inner third lateral boundary that extends behind and at an angle to the proximal edge of the platter.

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Background/Summary

BACKGROUND

(1) Bioptic indicia reader assemblies, such as those used in retail applications, can include high-end vision systems that include overhead cameras to provide expanded vision capabilities. However, overhead camera designs will not work with every installation, depending on the user's specific furniture layout and requirements. It would be beneficial to provide a bioptic indicia reader assembly that provides the expanded vision capabilities of a high-end vision system with overhead cameras with components that are contained within the footprint/volume of the standard bioptic indicia reader (i.e., without external cameras positioned above the bioptic indica reader).

SUMMARY

(2) In an embodiment, the present invention is a bioptic indicia reader assembly, including a housing and a platter. The housing has a lower housing portion, an upper housing portion extending above the lower housing portion, and a generally upright window positioned in the upper housing portion. The platter has a proximal edge adjacent the upper housing portion, a first lateral edge extending non-parallel to the proximal edge, a second lateral edge, opposite the first lateral edge, extending non-parallel to the proximal edge, a distal edge, opposite the proximal edge, extending non-parallel to the first and second lateral edges, a top surface facing a product scanning area, and a generally horizontal window positioned in the platter. The upper housing portion comprises a first protrusion located on a first side of a longitudinal centerline of the bioptic indicia reader assembly and extending forward of the generally upright window and towards the distal edge of the platter and a second protrusion located on a second side of the longitudinal centerline, opposite the first side, and extending forward of the generally upright window and towards the distal edge of the platter. The first protrusion comprises a first camera having a first field-of-view (FOV) having an inner first lateral boundary that extends behind and at an angle to the proximal edge of the platter and the second protrusion comprises a second camera having a second FOV having an inner third lateral boundary that extends behind and at an angle to the proximal edge of the platter.

(3) In a variation of this embodiment, the angle between the inner first lateral boundary of the first FOV and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.

(4) In another variation of this embodiment, the angle between the inner third lateral boundary and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.

(5) In another variation of this embodiment, the first FOV and the second FOV encompass at least 90% of a footprint of the platter.
(6) In another variation of this embodiment, the first FOV has an outer second lateral boundary, opposite the inner first lateral boundary, that extends outside of and at an angle to the first lateral edge of the platter.
(7) In another variation of this embodiment, the second FOV has an outer fourth lateral boundary, opposite the inner third lateral boundary, that extends outside of and at an angle to the second lateral edge of the platter.
(8) In another variation of this embodiment, a first upper boundary of the first FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
(9) In another variation of this embodiment, a second upper boundary of the second FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
(10) In another variation of this embodiment, the first protrusion comprises a third camera having a third FOV that is directed away from the first lateral edge of the platter and that has a third lower boundary that extends at an angle below the top surface of the platter.
(11) In another variation of this embodiment, the second protrusion comprises a fourth camera having a fourth FOV that is directed away from the second lateral edge of the platter and has a fourth lower boundary that extends at an angle below the top surface of the platter.
(12) In another embodiment, the present invention is a bioptic indicia reader assembly including a housing and a platter. The housing has a lower housing portion, an upper housing portion extending above the lower housing portion, and a generally upright window positioned in the upper housing portion. The platter has a proximal edge adjacent the upper housing portion, a first lateral edge extending non-parallel to the proximal edge, a second lateral edge, opposite the first lateral edge, extending non-parallel to the proximal edge, a distal edge, opposite the proximal edge, extending non-parallel to the first and second lateral edges, a top surface facing a product scanning area, and a generally horizontal window positioned in the platter. The upper housing portion comprises a first protrusion located on a first side of a longitudinal centerline of the bioptic indicia reader assembly and extending forward of the generally upright window and towards the distal edge of the platter and a second protrusion located on a second side of the longitudinal centerline, opposite the first side, and extending forward of the generally upright window and towards the distal edge of the platter. The first protrusion comprises a first camera having a first field-of-view (FOV) having an inner first lateral boundary and an outer second lateral boundary and the second protrusion comprises a second camera having a second FOV having an inner third lateral boundary and an outer fourth lateral boundary. The inner first lateral boundary and the inner third lateral boundary form an angle that is greater than or equal to 170 degrees and less than or equal to 190 degrees.
(13) In a variation of this embodiment, the inner first lateral boundary extends behind and at an angle to the proximal edge of the platter; and the inner third lateral boundary extends behind and at an angle to the proximal edge of the platter.
(14) In another variation of this embodiment, the angle between the inner first lateral boundary of the first FOV and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.
(15) In another variation of this embodiment, the angle between the inner third lateral boundary and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.
(16) In another variation of this embodiment, the first FOV and the second FOV encompass at least 90% of a footprint of the platter.
(17) In another variation of this embodiment, the outer second lateral boundary extends outside of and at an angle to the first lateral edge of the platter.

(18) In another variation of this embodiment, the outer fourth lateral boundary extends outside of and at an angle to the second lateral edge of the platter.
(19) In another variation of this embodiment, a first upper boundary of the first FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
(20) In another variation of this embodiment, a second upper boundary of the second FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
(21) In another variation of this embodiment, the first protrusion comprises a third camera having a third FOV that is directed away from the first lateral edge of the platter and that has a third lower boundary that extends at an angle below the top surface of the platter.
(22) In another variation of this embodiment, the second protrusion comprises a fourth camera having a fourth FOV that is directed away from the second lateral edge of the platter and has a fourth lower boundary that extends at an angle below the top surface of the platter.
(23) In another embodiment, the present invention is a vision reader assembly for use with a bioptic indicia reader, the vision reader assembly including a first housing and a second housing.
(24) The first housing is configured to be removably secured to a first side of an upper housing portion of the bioptic indicia reader assembly such that the first housing extends forward of a generally upright window of the upper housing portion and towards a distal edge of a platter of the bioptic indicia reader assembly and comprises a first camera having a first field-of-view (FOV) having an inner first lateral boundary that extends behind and at an angle to a proximal edge of the platter.
(25) The second housing is configured to be removably secured to a second side, opposite the first side, of the upper housing portion of the bioptic indicia reader assembly such that the second housing extends forward of the generally upright window and towards the distal edge and comprises a second camera having a second field-of-view (FOV) having an inner third lateral boundary that extends behind and at an angle to the proximal edge of the platter.
(26) In a variation of this embodiment, the angle between the inner first lateral boundary of the first FOV and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.
(27) In another variation of this embodiment, the angle between the inner third lateral boundary and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.
(28) In another variation of this embodiment, the first FOV and the second FOV encompass at least 90% of a footprint of the platter.
(29) In another variation of this embodiment, the first FOV has an outer second lateral boundary, opposite the inner first lateral boundary, that extends outside of and at an angle to a first lateral edge of the platter.
(30) In another variation of this embodiment, the second FOV has an outer fourth lateral boundary, opposite the inner third lateral boundary, that extends outside of and at an angle to a second lateral edge of the platter.
(31) In another variation of this embodiment, a first upper boundary of the first FOV and a top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
(32) In another variation of this embodiment, a second upper boundary of the second FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
(33) In another variation of this embodiment, the first housing comprises a third camera having a third FOV that is directed away from a first lateral edge of the platter and that has a third lower boundary that extends at an angle below a top surface of the platter.

(34) In another variation of this embodiment, the second housing comprises a fourth camera having a fourth FOV that is directed away from a second lateral edge of the platter and has a fourth lower boundary that extends at an angle below a top surface of the platter.

(35) In another embodiment, the present invention is a bioptic indicia reader assembly including a housing and a platter. The housing has a lower housing portion, an upper housing portion extending above the lower housing portion, and a generally upright window positioned in the upper housing portion. The platter has a proximal edge adjacent the upper housing portion, a first lateral edge extending non-parallel to the proximal edge, a second lateral edge, opposite the first lateral edge, extending non-parallel to the proximal edge, a distal edge, opposite the proximal edge, extending non-parallel to the first and second lateral edges, a top surface facing a product scanning area, and a generally horizontal window positioned in the platter. The upper housing portion comprises a first protrusion located on a first side of a longitudinal centerline of the bioptic indicia reader assembly and extending forward of the generally upright window and towards the distal edge of the platter and a second protrusion located on a second side of the longitudinal centerline, opposite the first side, and extending forward of the generally upright window and towards the distal edge of the platter. The first protrusion comprises a first camera having a first field-of-view (FOV) extending at least partially over the platter and covering at least 80% of an input area adjacent the bioptic indicia reader assembly. The second protrusion comprises a second camera having a second FOV extending at least partially over the platter.

(36) In a variation of this embodiment, the second FOV covers at least 80% of an output area adjacent the bioptic indicia reader assembly and opposite the input area.

(37) In another variation of this embodiment, the first FOV has an inner first lateral boundary that extends behind and at an angle to the proximal edge of the platter.

(38) In another variation of this embodiment, the angle between the inner first lateral boundary of the first FOV and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.

(39) In another variation of this embodiment, the second FOV has an inner third lateral boundary that extends behind and at an angle to the proximal edge of the platter.

(40) In another variation of this embodiment, the angle between the inner third lateral boundary and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.

(41) In another variation of this embodiment, an inner first lateral boundary of the first FOV and an inner third lateral boundary of the second FOV form an angle that is greater than or equal to 170 degrees and less than or equal to 190 degrees.

(42) In another variation of this embodiment, the first FOV and the second FOV encompass at least 90% of a footprint of the platter.

(43) In another variation of this embodiment, the first FOV has an outer second lateral boundary that extends outside of and at an angle to the first lateral edge of the platter.

(44) In another variation of this embodiment, the second FOV has an outer fourth lateral boundary that extends outside of and at an angle to the second lateral edge of the platter.

(45) In another variation of this embodiment, a first upper boundary of the first FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.

(46) In another variation of this embodiment, a second upper boundary of the second FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.

(47) In another variation of this embodiment, the first protrusion comprises a third camera having a third FOV that is directed away from the first lateral edge of the platter and that has a third lower boundary that extends at an angle below the top surface of the platter.

(48) In another variation of this embodiment, the second protrusion comprises a fourth camera

having a fourth FOV that is directed away from the second lateral edge of the platter and has a fourth lower boundary that extends at an angle below the top surface of the platter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.

(2) FIG. **1** illustrates a perspective view of a first example bioptic indicia reader assembly;

(3) FIG. **2** illustrates a side view of the bioptic indicia reader assembly of FIG. **1**, showing first and second reader FsOV;

(4) FIG. **3** illustrates a top view of the bioptic indicia reader assembly of FIG. **1**;

(5) FIG. **4** illustrates a side view of the bioptic indicia reader assembly of FIG. **1**, showing a first FOV of a first camera and a fifth FOV of a fifth camera;

(6) FIG. **5** illustrates a front view of the bioptic indicia reader assembly of FIG. **1**;

(7) FIG. **6** illustrates a top view of a second example bioptic indicia reader assembly;

(8) FIG. **7** illustrates a side view of the bioptic indicia reader assembly of FIG. **6**;

(9) FIG. **8** illustrates a front view of the bioptic indicia reader assembly of FIG. **6**;

(10) FIG. **9** illustrates a perspective view of an example vision reader assembly;

(11) FIG. **10** illustrates a top view of the vision reader assembly of FIG. **9** secured to an example bioptic indicia reader assembly;

(12) FIG. **11** illustrates a side view of the vision reader assembly and bioptic indicia reader assembly of FIG. **10**; and

(13) FIG. **12** illustrates a front view of the vision reader assembly and bioptic indicia reader assembly of FIG. **10**.

(14) Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity, have not necessarily been drawn to scale, and that details that are not necessary for an understanding of the invention or that render other details difficult to perceive may be omitted. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

(15) The apparatus components have been represented where appropriate by conventional symbols in the drawings, showing only those components and specific details that are pertinent to understanding the examples of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

(16) The example bioptic indicia reader assemblies disclosed herein provide expanded vision capabilities, similar to those provided by high-end vision systems with overhead cameras, with components that are contained within the footprint/volume of the bioptic indicia reader assembly (i.e., without external cameras positioned above the bioptic indica reader). This can allow users that have certain checkout arrangements that they do not want to change to upgrade to a bioptic indicia reader assembly with expanded vision capabilities without changing the checkout arrangement.

(17) The example bioptic indicia reader assemblies include protrusions on the bioptic indicia reader housing (or housings that can be mounted to an existing bioptic indicia reader housing) that each include one or more cameras. Locating multiple cameras in multiple protrusions in the housing (or housings mounted to the bioptic indicia reader housing) can keep the cameras within the

footprint/volume of the bioptic indicia reader, while providing a premium vision system completely contained within a form factor that can be dropped into any existing furniture solutions that current bioptic indicia reader assemblies can fit into. In some implementations, no cameras are positioned above the housing of the bioptic indicia reader assembly. In other implementations, the protrusions in the housing (or housings mounted to the bioptic indicia reader housing) can extend outside of the footprint/volume of the bioptic indicia reader if desired.

(18) The cameras in the protrusions (or housings that can be mounted to the bioptic indicia reader housing) can be positioned and configured such that the FsOV of the cameras cover: the entirety of the product scanning area (from multiple different angles); the face of a user (from multiple different angles); an input area (e.g., top edge of a cart or a length of an input conveyor); and/or an output area (e.g., top edge of a bag or a length of an output conveyor). To cover the face of a user of average height, an additional camera could also be centered in an upper housing portion and directed upward towards the face of the user. Additional cameras can also be provided in the protrusions (or housings that can be mounted to the bioptic indicia reader housing) that are directed outward to the side and downward from the upper housing portion (e.g., to see into a cart, into a lowered bagging area, etc.).

(19) Referring to FIGS. **1-5**, a first example bioptic indicia reader assembly **100**A is illustrated that can be configured to be supported by a workstation, such as a checkout counter at a point-of-sale (POS) of a retail store. In the example shown, bioptic indicia reader assembly **100**A is a bioptic barcode reader, but can be any type of indicia reader desired, such as a single window barcode reader, a radio-frequency identification reader, etc. Bioptic indicia reader assembly **100**A generally includes a housing **200** and a platter **500**. In some implementations, a perimeter frame (not shown) could at least partially surround and support bioptic indicia reader assembly **100**A, a sheet metal frame (not shown) could be secured to the perimeter frame, and a scale assembly (not shown) could be positioned between bioptic indicia reader assembly **100**A and the sheet metal frame to engage platter **500** to allow objects placed on platter **500** to be weighed by the scale assembly, if used. The sheet metal frame can be a single, unitary, integral part of can include multiple parts that are assembled together.

(20) Housing **200** has a lower housing portion **205** and an upper housing portion **225** extending above lower housing portion **205**. At least one generally upright window **230** can be positioned in upper housing portion **225** to allow a set of optical components positioned within housing **200** to direct at least a portion of a first reader FOV **125** through generally upright window **230**. In the implementation shown, bioptic indicia reader assembly **100**A includes two generally upright windows **230**, **230**A and intersecting FsOV extending through generally upright windows **230**, **230**A. However, in some implementations, bioptic indicia reader assembly **100**A can include a single generally upright window or more than two generally upright windows.

(21) Platter **500** has a proximal edge **505** that is adjacent upper housing portion **225**. A first lateral edge **510** extends non-parallel (perpendicular in implementation shown) to proximal edge **505** and a second lateral edge **515**, opposite first lateral edge **510**, extends non-parallel (perpendicular in implementation shown) to proximal edge **505**. A distal edge **520**, opposite proximal edge **505**, extends non-parallel (perpendicular in implementation shown) to first lateral edge **510** and second lateral edge **515**. A generally horizontal window **535** can be positioned in platter **500** to allow the set of optical components to direct at least a portion of a second reader field-of-view **130** through generally horizontal window **535**. First reader FOV **125** directed through generally upright window **230** (and the FOV directed through generally upright window **230**A if used) and second reader field-of-view **130** directed through generally horizontal window **535** intersect to define a product scanning area **105**, where an object can be scanned for sale at the POS. A top surface **530** of platter **500** faces a product scanning area **105**.

(22) Upper housing portion **225** has a first protrusion **300** located on a first side **235** of a longitudinal centerline **540** of bioptic indicia reader assembly **100**A (left side in orientation shown

if FIG. **3**) and a second protrusion **400** located on a second side **240** of longitudinal centerline **540** (right side in orientation shown in FIG. **3**), opposite first side **235**. First protrusion **300** and second protrusion **400** each extend forward of generally upright window **230** and generally upright window **230**A, towards distal edge **520** of platter **500** from upper housing portion **225**, which can allow optical elements positioned within first protrusion **300** and/or second protrusion **400** to see past a front surface **265** of upper housing portion **225**, where generally upright window **230** and generally upright window **230**A are located.

(23) First protrusion **300** in upper housing portion **225** can include a first camera **305** that has a first FOV **310** that extends at least partially over platter **500** and covers at least 80% of an input area **115** adjacent platter **500** of bioptic indicia reader assembly **100**A. As used herein, input area **115** is an area adjacent bioptic indicia reader assembly **100**A where a user can position a cart and/or place items to be scanned, or where an input conveyor directs items toward bioptic indicia reader assembly **100**A. Input area **115** can be located on either side of bioptic indicia reader assembly **100**A and can have a depth D**1** that is approximately the same as the distance from proximal edge **505** of platter **500** to distal edge **520** and a length L**1** that extends approximately 2 feet from second lateral edge **515** (or first lateral edge **510** depending on the particular setup) of platter **500**. In the particular implementation shown, an inner first lateral boundary **315** of first FOV **310** extends behind (i.e., at an angle away from proximal edge **505** of platter **500** and towards upper housing portion **225**) and at an angle **320** to proximal edge **505** of platter **500**, which can allow first FOV **310** to substantially cover input area **115**. Angle **320** between inner first lateral boundary **315** and proximal edge **505** can be greater than or equal to 1 degree and less than or equal to 25 degrees. In the implementation shown, angle **320** is 10 degrees, which allows first FOV **310** to cover all of input area **115**. First FOV **310** also has an outer second lateral boundary **325**, opposite inner first lateral boundary **315**, that extends outside of and at an angle **330** to first lateral edge **510** of platter **500**. Angle **330** between outer second lateral boundary **325** and first lateral edge **510** can be greater than or equal to 0 degrees and less than or equal to 5 degrees, which can provide substantial coverage of product scanning area **105** by first FOV **310**. First FOV **310** also has a first upper boundary **335** that forms an angle **340** with top surface **530** of platter **500**. Angle **340** can be greater than or equal to 50 degrees and less than or equal to 100 degrees. In the implementation shown, angle **340** is 60 degrees, which can allow first FOV **310** to encompass the face of an average height user.

(24) First camera **305** can also be configured to identify a predetermined illumination, such as a light of a certain color, brightness, and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through first FOV **310** have been properly scanned before they are placed in output area **120**.

(25) First protrusion **300** can also include a third camera **345** that has a third FOV **350** that is directed away from first lateral edge **510** of platter **500**. A third lower boundary **355** of third FOV **350** can extend at an angle **360** below top surface **530** of platter **500**, which can allow third FOV **350** to see into a cart positioned in input area **115** or to see products placed in input area **115**.

(26) Second protrusion **400** in upper housing portion **225** includes a second camera **405** that has a second FOV **410** that extends at least partially over platter **500** and covers at least 80% of an output area **120**, opposite input area **115**, adjacent platter **500** of bioptic indicia reader assembly **100**A. As used herein, output area **120** is an area adjacent bioptic indicia reader assembly **100**A where a user can place/bag items already scanned, or where an output conveyor directs items away from bioptic indicia reader assembly **100**A. Output area **120** can be located on either side of bioptic indicia reader assembly **100**A, opposite input area **115**, and can have a depth D**2** that is approximately the same as the distance from proximal edge **505** of platter **500** to distal edge **520** and a length L**2** that extends approximately 2 feet from first lateral edge **510** (or second lateral edge **515** depending on the particular setup) of platter **500**. In the particular implementation shown, an inner third lateral boundary **415** of second FOV **410** extends behind (i.e., at an angle away from proximal edge **505** of

platter **500** and towards upper housing portion **225**) and at an angle **420** to proximal edge **505** of platter **500**. Angle **420** between inner third lateral boundary **415** and proximal edge **505** can be greater than or equal to 1 degree and less than or equal to 25 degrees. In the implementation shown, angle **420** is 10 degrees, which allows second FOV **410** to cover all of output area **120**. Second FOV **410** also has an outer fourth lateral boundary **430**, opposite inner third lateral boundary **415**, that extends outside of an at an angle **435** to second lateral edge **515** of platter **500**. Angle **435** between outer fourth lateral boundary **430** and second lateral edge **515** can be greater than or equal to 0 degrees and less than or equal to 5 degrees, which can provide substantial coverage of product scanning area **105** by second FOV **410**. Second FOV **410** also has a second upper boundary **440** that forms an angle **445** with top surface **530** of platter **500**. Angle **445** can be greater than or equal to 50 degrees and less than or equal to 100 degrees. In the implementation shown, angle **445** is 60 degrees, which can allow second FOV **410** to encompass the face of an average height user.
(27) Second camera **405** can also be configured to identify a predetermined illumination, such as a light of a certain color, brightness, and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through second FOV **410** have been properly scanned before they are placed in output area **120**.
(28) Second protrusion **400** can also include a fourth camera **450** that has a fourth FOV **455** that is directed away from second lateral edge **515** of platter **500**. A fourth lower boundary **460** of fourth FOV **455** can extend at an angle **465** below top surface **530** of platter **500**, which can allow fourth FOV **455** to see into products/bags in output area **120**.
(29) In some implementations, first FOV **310** and second FOV **410** can encompass at least 90% of a footprint **525** of platter **500**, which ensures coverage of platter **500** and product scanning area **105** by first FOV **310** and second FOV **410**. As used herein footprint **525** of platter **500** is the outer perimeter of platter **500** when looking straight down from above bioptic indicia reader assembly **100**A.
(30) In some implementations, upper housing portion **225** can also include a fifth camera **245** that can be positioned in upper housing portion **225** and proximate a longitudinal centerline **540** of bioptic indicia reader assembly **100**A. As used herein, proximate longitudinal centerline **540** means within 1 inch of longitudinal centerline **540** in any direction. Fifth camera **245** can have a fifth FOV **250** that extends over platter **500**. A fifth upper boundary **255** of fifth FOV **250** can form an angle **260** with top surface **530** of platter **500** that is greater than or equal to 50 degrees and less than or equal to 100 degrees. In the implementation shown, angle **260** is approximately 60 degrees, which can allow fifth FOV **250** to encompass the face of an average height user.
(31) In some implementations, bioptic indicia reader assembly **100**A can also include a microphone (not shown), or other audio detector, that is positioned at least partially within housing **200** and that can be configured to identify a predetermined audio signal, such as a sound of a certain tone, loudness, and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through first FOV **310** and/or second FOV **410** have been properly scanned before they are placed in output area **120**.
(32) Referring to FIGS. **6-8**, a second example bioptic indicia reader assembly **100**B is illustrated, which generally includes housing **200** and platter **500**, as described above for bioptic indicia reader assembly **100**A.
(33) In bioptic indicia reader assembly **1001**B, first protrusion **300** in upper housing portion **225** can include a first camera **305**A that has a first FOV **310**A that extends at least partially over platter **500** and covers at least 80% of input area **115** adjacent platter **500** of bioptic indicia reader assembly **100**B. First FOV **310**A has an inner first lateral boundary **315**A and an outer second lateral boundary **325**A, opposite inner first lateral boundary **315**A. In the particular implementation shown, inner first lateral boundary **315**A of first FOV **310**A can extend in front of (i.e., at an angle towards proximal edge **505** of platter **500** and away from upper housing portion **225**) and at an angle **320**A to proximal edge **505** of platter **500**, but could also extend behind and at angle **320** to

proximal edge **505** of platter **500**, as described above for first FOV **310**. Angle **320**A between inner first lateral boundary **315**A and proximal edge **505** can be greater than or equal to 1 degree and less than or equal to 25 degrees. In the implementation shown, angle **320**A is 10 degrees, which allows first FOV **310** to cover at least 80% of input area **115**. Outer second lateral boundary **325**A extends outside of and at angle **330**A to first lateral edge **510** of platter **500**. Angle **330**A between outer second lateral boundary **325**A and first lateral edge **510** can be greater than or equal to 0 degrees and less than or equal to 5 degrees, which can provide substantial coverage of product scanning area **105** by first FOV **310**A. First FOV **310**A also has a first upper boundary **335**A that forms an angle **340**A with top surface **530** of platter **500**. Angle **340**A can be greater than or equal to 50 degrees and less than or equal to 100 degrees. In the implementation shown, angle **340**A is 60 degrees, which can allow first FOV **310**A to encompass the face of an average height user.
(34) First camera **305**A can also be configured to identify a predetermined illumination, such as a light of a certain color, brightness, and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through first FOV **310**A have been properly scanned before they are placed in output area **120**.
(35) First protrusion **300** can also include third camera **345** that has third FOV **350**, as described above for bioptic indicia reader assembly **100**A.
(36) In bioptic indicia reader assembly **100**B, second protrusion **400** in upper housing portion **225** includes a second camera **405**A that has a second FOV **410**A that extends at least partially over platter **500** and covers at least 80% of output area **120**, opposite input area **115**, adjacent platter **500** of bioptic indicia reader assembly **100**B. Second FOV **410**A has an inner third lateral boundary **415**A and an outer fourth lateral boundary **430**A. In the particular implementation shown, inner third lateral boundary **415**A extends in front of (i.e., at an angle towards proximal edge **505** of platter **500** and away from upper housing portion **225**) and at an angle **420**A to proximal edge **505** of platter **500**, but could also extend behind and at angle **420** to proximal edge **505** of platter **500**, as described above for second FOV **410**. Angle **420**A between inner third lateral boundary **415**A and proximal edge **505** can be greater than or equal to 1 degree and less than or equal to 25 degrees. In the implementation shown, angle **420**A is 10 degrees, which allows second FOV **410**A to cover all of output area **120**. Inner first lateral boundary **315**A of first FOV **310**A and inner third lateral boundary **415**A of second FOV **410**A form an angle **425** that is greater than or equal to 170 degrees and less than or equal to 190 degrees. Outer fourth lateral boundary **430**A extends outside of and at angle **435**A to second lateral edge **515** of platter **500**. Angle **435**A between outer fourth lateral boundary **430**A and second lateral edge **515** can be greater than or equal to 0 degrees and less than or equal to 5 degrees, which can provide substantial coverage of product scanning area **105** by second FOV **410**A. Second FOV **410**A also has a second upper boundary **440**A that forms an angle **445**A with top surface **530** of platter **500**. Angle **445**A can be greater than or equal to 50 degrees and less than or equal to 100 degrees. In the implementation shown, angle **445**A is 60 degrees, which can allow second FOV **410**A to encompass the face of an average height user.
(37) Second camera **405**A can also be configured to identify a predetermined illumination, such as a light of a certain color, brightness, and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through second FOV **410**A have been properly scanned before they are placed in output area **120**.
(38) Second protrusion **400** can also include fourth camera **450** that has fourth FOV **455**, as described above for bioptic indicia reader assembly **100**A.
(39) In some implementations, first FOV **310**A and second FOV **410**A encompass at least 90% of footprint **525** of platter **500** which ensures coverage of platter **500** and product scanning area **105** by first FOV **310**A and second FOV **410**A.
(40) In some implementations, upper housing portion **225** can also include fifth camera **245** having fifth FOV **250**, as described above for bioptic indicia reader assembly **100**A.
(41) In some implementations, bioptic indicia reader assembly **100**B can also include a microphone

(not shown), or other audio detector, that is positioned at least partially within housing **200** and that can be configured to identify a predetermined audio signal, such as a sound of a certain tone, loudness, and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through first FOV **310**A and/or second FOV **410**A have been properly scanned before they are placed in output area **120**.

(42) Referring to FIG. **9**, an example vision reader assembly **110** is illustrated. FIGS. **10-12** illustrate vision reader assembly **110** secured to an example bioptic indicia reader assembly **110**C. In the implementation shown in FIGS. **10-12**, bioptic indicia reader assembly **110**C is a bioptic barcode reader, but can be any type of indicia reader desired, such as a single window barcode reader, a radio-frequency identification reader, etc. Bioptic indicia reader assembly **1000** generally includes a housing **600** (corresponding to housing **200**) and platter **500**, as described above. In some implementations, a perimeter frame (not shown) could at least partially surround and support bioptic indicia reader assembly **1000**, a sheet metal frame (not shown) could be secured to the perimeter frame, and a scale assembly (not shown) could be positioned between bioptic indicia reader assembly **1000** and the sheet metal frame to engage platter **500** to allow objects placed on platter **500** to be weighed by the scale assembly, if used. The sheet metal frame can be a single, unitary, integral part of can include multiple parts that are assembled together.

(43) Housing **600** has a lower housing portion **605** and an upper housing portion **625** extending above lower housing portion **605**. As discussed above for housing **200**, at least one generally upright window **630** can be positioned in upper housing portion **625** to allow a set of optical components positioned within housing **600** to direct at least a portion of first reader FOV **125** through generally upright window **630**. In the implementation shown, housing **600** of bioptic indicia reader assembly **1000** includes two generally upright windows **630**, **630**A and intersecting FsOV extending through generally upright windows **630**, **630**A. However, in some implementations, housing **600** of bioptic indicia reader assembly **1000** can include a single generally upright window or more than two generally upright windows. As described above, first reader FOV **125** directed through generally upright window **630**, the FOV directed through generally upright window **630**A, and second reader field-of-view **130** directed through generally horizontal window **535** of platter **500** intersect to define a product scanning area **105**, where an object can be scanned for sale at the POS.

(44) Vision reader assembly **110** generally includes a first housing **700** and a second housing **800**. First housing **700** is configured to be removably secured to a first side **635** of upper housing portion **625** of housing **600** and second housing **800** is configured to be removably secured to a second side **640** of upper housing portion **625** of housing **600**, such that first housing **700** and second housing **800** each extend forward of generally upright window **630** and towards distal edge **520** of platter **500**. In some implementations, first housing **700** and second housing **800** can be removably secured directly to upper housing portion, for example, through mechanical means, such as screws or hooks, through a hook and loop type fastener, via adhesives, etc. In other implementations, first housing **700** and second housing **800** can be mounted to a sleeve (e.g., a generally U-shaped sleeve) that can be removably secured over upper housing portion **625**. In these implementations, control electronics can also be located within the sleeve.

(45) First housing **700** can include a first camera **705** that has a first FOV **710** that extends at least partially over platter **500** and covers at least 80% of input area **115** adjacent platter **500** of bioptic indicia reader assembly **1000**, with first housing **700** secured to upper housing portion **625**, as described above for first FOV **310**. In the particular implementation shown, an inner first lateral boundary **715** of first FOV **710** extends behind (i.e., at an angle away from proximal edge **505** of platter **500** and towards upper housing portion **225**) and at an angle **720** to proximal edge **505** of platter **500**, which can allow first FOV **710** to substantially cover input area **115**. Angle **720** between inner first lateral boundary **715** and proximal edge **505** of platter **500** can be greater than or equal to 1 degree and less than or equal to 25 degrees. In the implementation shown, angle **720**

is 10 degrees, which allows first FOV **710** to cover all of input area **115**. First FOV **710** also has an outer second lateral boundary **725**, opposite inner first lateral boundary **715**, that extends outside of an at an angle **730** to first lateral edge **510** of platter **500**. Angle **730** between outer second lateral boundary **725** and first lateral edge **510** can be greater than or equal to 0 degrees and less than or equal to 5 degrees, which can provide substantial coverage of a product scanning area **105** by first FOV **710**. First FOV **710** also has a first upper boundary **735** that forms an angle **740** with top surface **530** of platter **500**. Angle **740** can be greater than or equal to 50 degrees and less than or equal to 100 degrees. In the implementation shown, angle **740** is 60 degrees, which can allow first FOV **710** to encompass the face of an average height user.
(46) First camera **705** can also be configured to identify a predetermined illumination, such as a light of a certain color, brightness, and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through first FOV **710** have been properly scanned before they are placed in output area **120**.
(47) First housing **700** can also include a third camera **745** that has a third FOV **750** that is directed away from first lateral edge **510** of platter **500**. A third lower boundary **755** of third FOV **750** extends at an angle **760** below top surface **530** of platter **500**, which can allow third FOV **750** to see into a cart positioned in input area **115** or to see products placed in input area **115**.
(48) Second housing **800** can include a second camera **805** that has a second FOV **810** that extends at least partially over platter **500** and covers at least 80% of output area **120** adjacent platter **500**, opposite input area **115**, with second housing **800** secured to upper housing portion **625**, as described above for second FOV **410**. In the particular implementation shown, an inner third lateral boundary **815** of second FOV **810** extends behind (i.e., at an angle away from proximal edge **505** of platter **500** and towards upper housing portion **225**) and at an angle **820** to proximal edge **505** of platter **500**. Angle **820** between inner third lateral boundary **815** and proximal edge **505** of platter **500** can be greater than or equal to 1 degree and less than or equal to 25 degrees. In the implementation shown, angle **820** is 10 degrees, which allows second FOV **810** to cover all of output area **120**. Second FOV **810** also has an outer fourth lateral boundary **830**, opposite inner third lateral boundary **815**, that extends outside of and at an angle **835** to second lateral edge **515** of platter **500**. Angle **835** between outer fourth lateral boundary **830** and second lateral edge **515** can be greater than or equal to 0 degrees and less than or equal to 5 degrees, which can provide substantial coverage of product scanning area **105** by second FOV **810**. Second FOV **810** also has a second upper boundary **840** that forms an angle **845** with top surface **530** of platter **500**. Angle **845** can be greater than or equal to 50 degrees and less than or equal to 100 degrees. In the implementation shown, angle **845** is 60 degrees, which can allow second FOV **810** to encompass the face of an average height user.
(49) Second camera **805** can also be configured to identify a predetermined illumination, such as a light of a certain color, brightness, and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through second FOV **810** have been properly scanned before they are placed in output area **120**.
(50) Second housing **800** can also include a fourth camera **850** that has a fourth FOV **855** that is directed away from second lateral edge **515** of platter **500**. A fourth lower boundary **860** of fourth FOV **855** extends at an angle **865** below top surface **530** of platter **500**, which can allow fourth FOV **855** to see into products/bags in output area **120**.
(51) In some implementations, first FOV **710** and second FOV **810** can encompass at least 90% of footprint **525** of platter **500**, with first housing **700** and second housing **800** secured to upper housing portion **625** of bioptic indicia reader assembly **1000**, which ensures coverage of platter **500** and product scanning area **105** by first FOV **710** and second FOV **810**.
(52) In some implementations, vision reader assembly **110** can also include a microphone (not shown) (e.g., in first housing **700** and/or second housing **800**), or other audio detector, that can be configured to identify a predetermined audio signal, such as a sound of a certain tone, loudness,

and/or duration, that indicates a good scan of a barcode on a product, which can be used to verify that all products passed through first FOV **710** and/or second FOV **810** have been properly scanned before they are placed in output area **120**.
(53) In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings. Additionally, the described embodiments/examples/implementations should not be interpreted as mutually exclusive, and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned embodiments/examples/implementations may be included in any of the other aforementioned embodiments/examples/implementations.
(54) The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.
(55) Moreover, in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms "comprises," "comprising," "has", "having," "includes", "including," "contains", "containing" or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element proceeded by "comprises . . . a", "has . . . a", "includes . . . a", "contains . . . a" does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms "a" and "an" are defined as one or more unless explicitly stated otherwise herein. The terms "substantially", "essentially", "approximately", "about" or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term "coupled" as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is "configured" in a certain way is configured in at least that way, but may also be configured in ways that are not listed.
(56) The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A bioptic indicia reader assembly, comprising: a housing having a lower housing portion, an upper housing portion extending above the lower housing portion, and a generally upright window positioned in the upper housing portion; and a platter having a proximal edge adjacent the upper housing portion, a first lateral edge extending non-parallel to the proximal edge, a second lateral edge, opposite the first lateral edge, extending non-parallel to the proximal edge, a distal edge, opposite the proximal edge, extending non-parallel to the first and second lateral edges, a top surface facing a product scanning area, and a generally horizontal window positioned in the platter; wherein the upper housing portion comprises: a first protrusion located on a first side of a longitudinal centerline of the bioptic indicia reader assembly and extending forward of the generally upright window and towards the distal edge of the platter; and a second protrusion located on a second side of the longitudinal centerline, opposite the first side, and extending forward of the generally upright window and towards the distal edge of the platter; the first protrusion comprises a first camera having a first field-of-view (FOV) having an inner first lateral boundary that extends behind and at an angle to the proximal edge of the platter; and the second protrusion comprises a second camera having a second FOV having an inner third lateral boundary that extends behind and at an angle to the proximal edge of the platter.
2. The bioptic indicia reader assembly of claim 1, wherein the angle between the inner first lateral boundary of the first FOV and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.
3. The bioptic indicia reader assembly of claim 2, wherein the angle between the inner third lateral boundary and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.
4. The bioptic indicia reader assembly of claim 1, wherein the first FOV and the second FOV encompass at least 90% of a footprint of the platter.
5. The bioptic indicia reader assembly of claim 1, wherein the first FOV has an outer second lateral boundary, opposite the inner first lateral boundary, that extends outside of and at an angle to the first lateral edge of the platter.
6. The bioptic indicia reader assembly of claim 5, wherein the second FOV has an outer fourth lateral boundary, opposite the inner third lateral boundary, that extends outside of and at an angle to the second lateral edge of the platter.
7. The bioptic indicia reader assembly of claim 1, wherein a first upper boundary of the first FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
8. The bioptic indicia reader assembly of claim 7, wherein a second upper boundary of the second FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
9. The bioptic indicia reader assembly of claim 1, wherein the first protrusion comprises a third camera having a third FOV that is directed away from the first lateral edge of the platter and that has a third lower boundary that extends at an angle below the top surface of the platter.
10. The bioptic indicia reader assembly of claim 1, wherein the second protrusion comprises a fourth camera having a fourth FOV that is directed away from the second lateral edge of the platter and has a fourth lower boundary that extends at an angle below the top surface of the platter.
11. A bioptic indicia reader assembly, comprising: a housing having a lower housing portion, an upper housing portion extending above the lower housing portion, and a generally upright window positioned in the upper housing portion; and a platter having a proximal edge adjacent the upper housing portion, a first lateral edge extending non-parallel to the proximal edge, a second lateral edge, opposite the first lateral edge, extending non-parallel to the proximal edge, a distal edge, opposite the proximal edge, extending non-parallel to the first and second lateral edges, a top surface facing a product scanning area, and a generally horizontal window positioned in the platter;

wherein the upper housing portion comprises: a first protrusion located on a first side of a longitudinal centerline of the bioptic indicia reader and extending forward of the generally upright window and towards the distal edge of the platter; and a second protrusion located on a second side of the longitudinal centerline, opposite the first side, and extending forward of the generally upright window and towards the distal edge of the platter; the first protrusion comprises a first camera having a first field-of-view (FOV) having an inner first lateral boundary and an outer second lateral boundary; the second protrusion comprises a second camera having a second FOV having an inner third lateral boundary and an outer fourth lateral boundary; and the inner first lateral boundary and the inner third lateral boundary form an angle that is greater than or equal to 170 degrees and less than or equal to 190 degrees.
12. The bioptic indicia reader assembly of claim 11, wherein: the inner first lateral boundary extends behind and at an angle to the proximal edge of the platter; and the inner third lateral boundary extends behind and at an angle to the proximal edge of the platter.
13. The bioptic indicia reader assembly of claim 12, wherein the angle between the inner first lateral boundary of the first FOV and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.
14. The bioptic indicia reader assembly of claim 13, wherein the angle between the inner third lateral boundary and the proximal edge of the platter is greater than or equal to 1 degree and less than or equal to 25 degrees.
15. The bioptic indicia reader assembly of claim 11, wherein the first FOV and the second FOV encompass at least 90% of a footprint of the platter.
16. The bioptic indicia reader assembly of claim 11, wherein the outer second lateral boundary extends outside of and at an angle to the first lateral edge of the platter.
17. The bioptic indicia reader assembly of claim 16, wherein the outer fourth lateral boundary extends outside of and at an angle to the second lateral edge of the platter.
18. The bioptic indicia reader assembly of claim 11, wherein a first upper boundary of the first FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
19. The bioptic indicia reader assembly of claim 18, wherein a second upper boundary of the second FOV and the top surface of the platter form an angle that is greater than or equal to 50 degrees and less than or equal to 100 degrees.
20. The bioptic indicia reader assembly of claim 11, wherein the first protrusion comprises a third camera having a third FOV that is directed away from the first lateral edge of the platter and that has a third lower boundary that extends at an angle below the top surface of the platter.
21. The bioptic indicia reader assembly of claim 11, wherein the second protrusion comprises a fourth camera having a fourth FOV that is directed away from the second lateral edge of the platter and has a fourth lower boundary that extends at an angle below the top surface of the platter.